The Window Method of FIR Filter Design

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Key Concepts

1) The window method designs linear phase FIR filters by applying a window to an ideal filter impulse response to make it finite duration.

    a) The ideal filter impulse response is computed as the inverse DTFT of the desired frequency response.

    b) The ideal filter is usually assumed to have zero phase.

    c) The window is centered on the origin to match the zero phase response of the ideal filter.

    d) The designed filter coefficients are time shifted to make the filter causal.

2) A rectangular window minimizes the integral of the squared error between the desired and actual frequency response.

3) The designed filter response is the convolution of the window DTFT with the desired frequency response.

    a) The width of the filter transition bands is proportional to the main lobe width of the window.

    b) The pass and stop band ripple is proportional to the window side lobes.

    c) Different windows are used to trade transition bandwidth for pass/stop band ripple.

4) The Kaiser window approximately optimizes the tradeoff between main lobes and side lobes.

5) The window design method is strongly dependent on guess-and-check to find a filter that satisfies the specifications. The resulting design is not guaranteed to be optimal.

Video Screenshots

Assume linear phase, M even

$$H(e^{j\omega}) = \sum_{k=0}^{M} h[k]e^{-jk\omega} = e^{-j\omega M/2}A(\omega)$$

$A(\omega)$ ← real

Consider minimizing mean squared error

$$\min_{h[k]} \frac{1}{2\pi}\int_{-\pi}^{\pi} |H_d(e^{j\omega}) - H(e^{j\omega})|^2 d\omega$$

$H_d(e^{j\omega})$ desired response

Let $H_d(e^{j\omega}) = e^{-j\omega M/2}A_d(\omega)$

$$\min_{a[k]} \frac{1}{2\pi}\int_{-\pi}^{\pi} |A_d(\omega) - A(\omega)|^2 d\omega \quad ; \quad h[k+M/2] = a[k] \xleftrightarrow{DTFT} A(\omega)$$

Parsevals Theorem

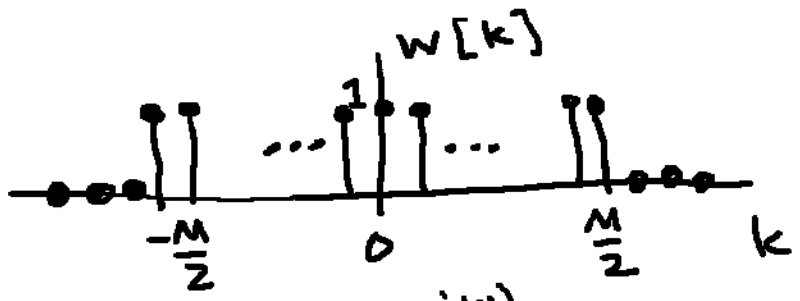
$$\min_{a[k]} \sum_{k=-\infty}^{\infty} |a_d[k] - a[k]|^2 \quad ; \quad a_d[k] \xleftrightarrow{DTFT} A_d(\omega)$$

$a[k]$ nonzero, $-\frac{M}{2} \le k \le \frac{M}{2}$

$$\min_{a[k]} \sum_{k=-\frac{M}{2}}^{M/2} |a_d[k] - a[k]|^2 + \sum_{k=-\infty}^{-M/2-1} |a_d[k]|^2 + \sum_{k=M/2+1}^{\infty} |a_d[k]|^2$$

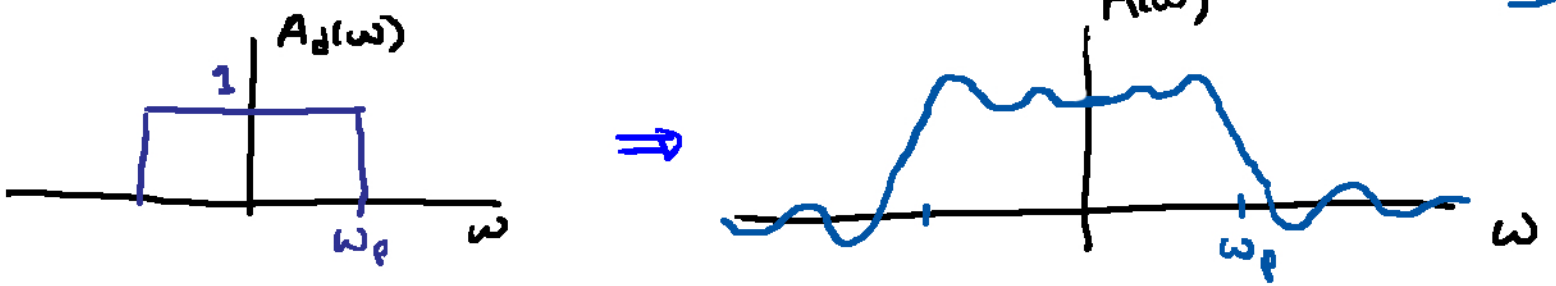
$$\Rightarrow \quad a[k] = a_d[k] \;, \quad -\frac{M}{2} \le k \le \frac{M}{2}$$

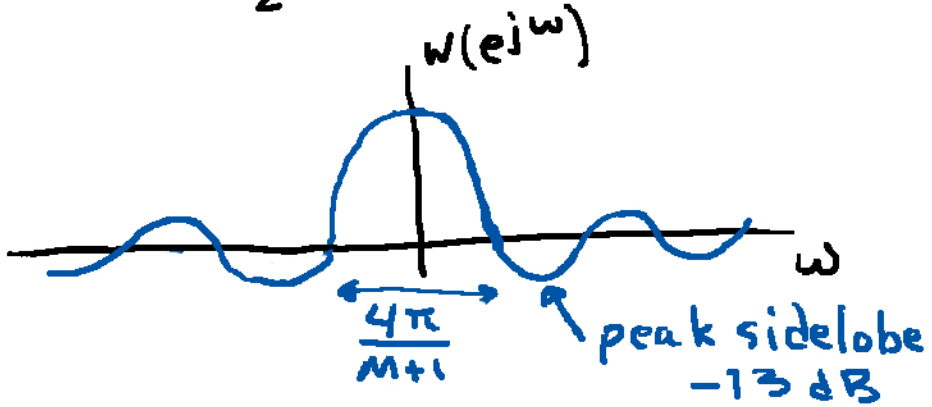
$$= a_d[k]\, w[k]$$



Nature of error $A_d(\omega) - A(\omega)$:

$$A(\omega) = A_d(\omega) * W(e^{j\omega})$$





Mainlobe width of $W(e^{j\omega})$ ⟶ transition bandwidth

Sidelobes of $W(e^{j\omega})$ ⟶ passband & stopband ripple

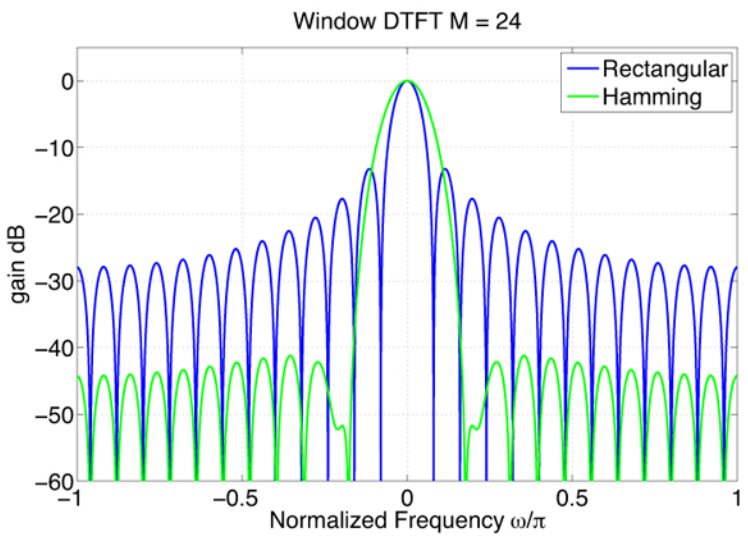
- -13 dB sidelobes of rectangular window ⇒ unacceptable ripple

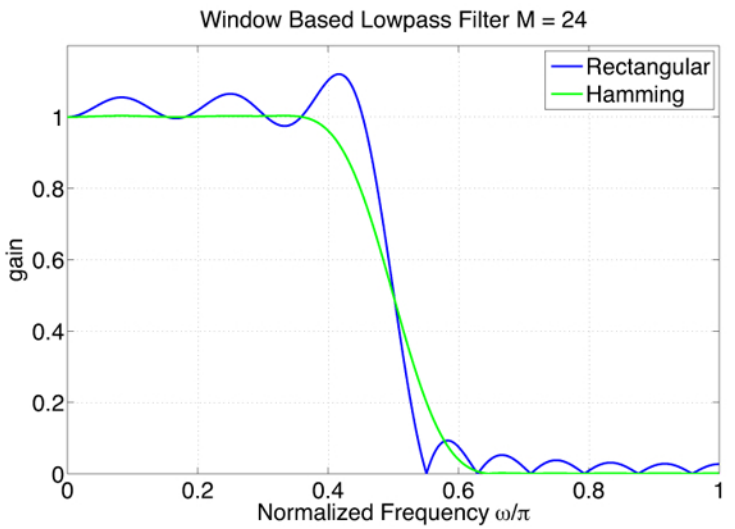
- Apply other windows to trade mainlobe width (transition band) for sidelobes (ripple)

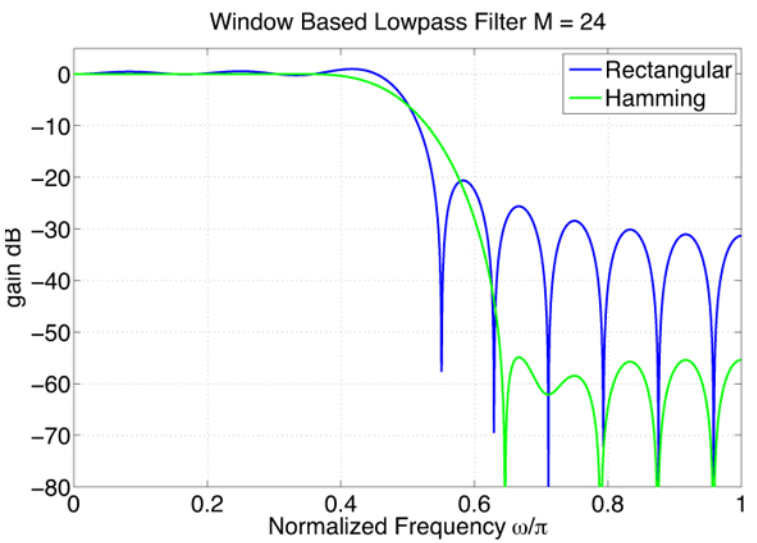
Hamming window: $w[k] = \begin{cases} 0.54 - 0.46\cos\left(2\pi\left(k+\frac{M}{2}\right)\right), & -\frac{M}{2} \le k \le \frac{M}{2} \\ 0, & \text{other} \end{cases}$

Mainlobe width: $8\pi/M$

Peak sidelobe height: -41 dB







Kaiser window: approximately optimize mainlobe sidelobe tradeoff

parameters β, M :

$$w[n] = \begin{cases} \dfrac{I_0\left[\beta\left(1-\left(\frac{2n}{M}\right)^2\right)^{1/2}\right]}{I_0(\beta)}, & -\frac{M}{2} \le n \le \frac{M}{2} \\ 0, & \text{otherwise} \end{cases}$$

$I_0(\cdot)$: zeroth order modified Bessel function of 1ST kind

$\beta = 0 \Rightarrow$ rectangular window

$\beta \uparrow \Rightarrow$ mainlobe width $\uparrow$, sidelobe height $\downarrow$

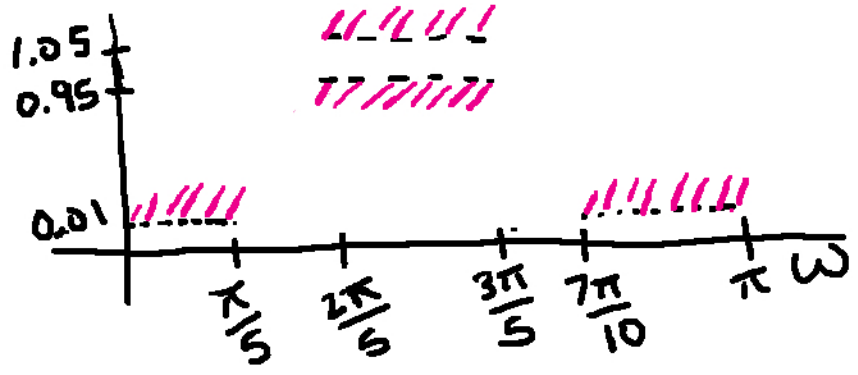
$\alpha > 0$ dB max error :

$$\beta = \begin{cases} 0.1102(\alpha - 8.7) & \alpha > 50 \\ 0.5842(\alpha-21)^{0.4} + 0.07886(\alpha-21) & 21 \le \alpha \le 50 \\ 0 & \alpha < 21 \end{cases}$$

Filter order $M = (\alpha - 8)/(2.285\Delta\omega)$

$\Delta\omega$: transition bandwidth

Example: Bandpass filter



passband error < 0.05
stopband error < 0.01
transition band $\approx \pi/10$

```
F = [ pi/5,   2*pi/5,   3*pi/5,   7*pi/10]/pi;
A = [ 0   1   0 ];
Dev = [ 0.01   0.05   0.01];

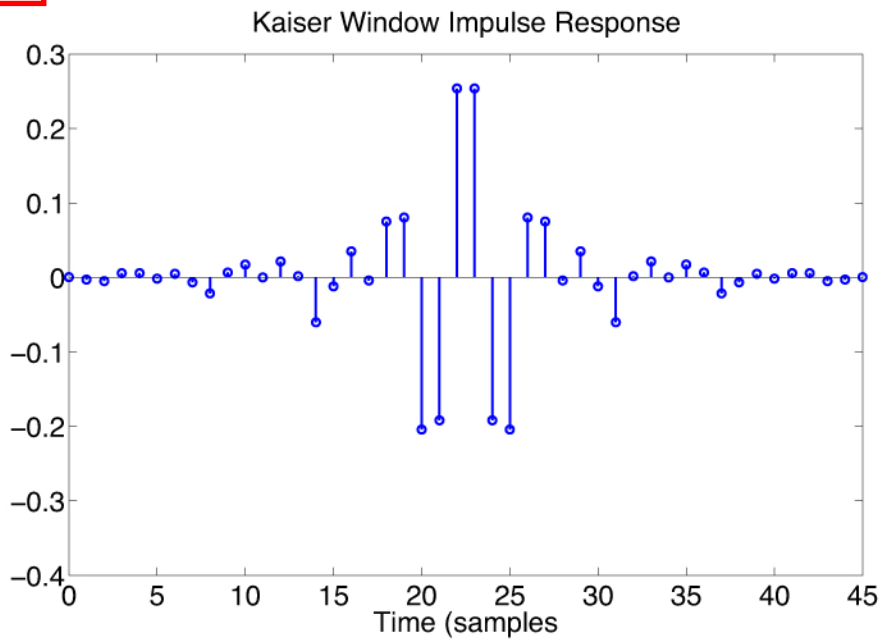
[M, Wn, beta, filtype] = kaiserord(F, A, Dev);
b = fir1(M, Wn, filtype, kaiser(M+1,beta), 'noscale');
```

$M = 45$

$\beta = 3.3953$

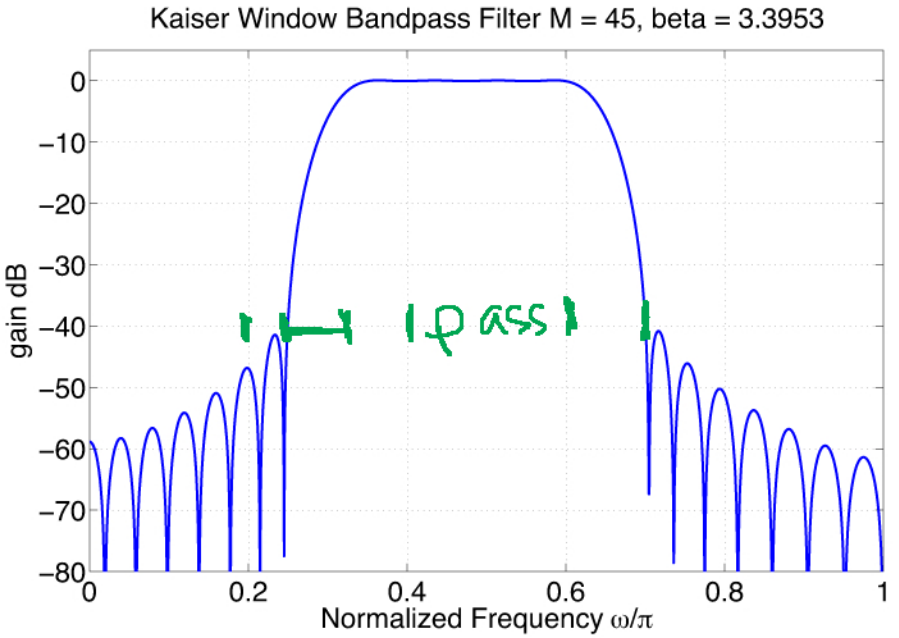
Kaiser Window Bandpass Filter

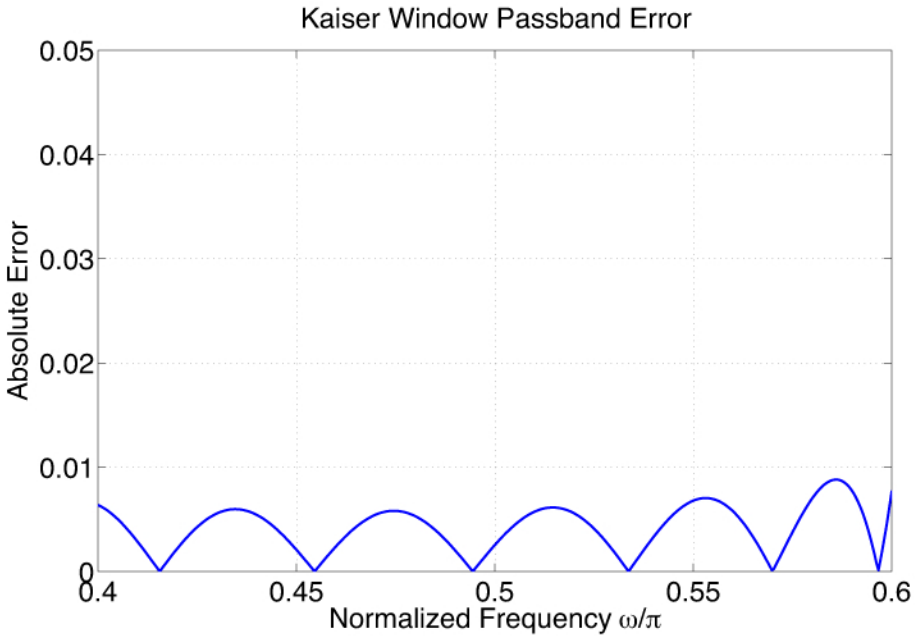
13:35



passband error < 0.05

Stopband error < 0.01





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